CHAPTER 11

COMPUTER NETWORKS

CHAPTER ESSENCE AT A GLANCE:
Master the architectural framework of distributed communications spanning physical cabling, wireless media, network topologies, routing hardware, and the layered OSI protocol stack.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Network Geographical Scale:** Networks scale from short-range personal area networks (PAN) and local networks (LAN) up to metropolitan (MAN) and globally interconnected wide area networks (WAN).

• **Guided versus Wireless Transmission Physics:** Guided copper cables transmit electrical signals and fiber optics transmit light pulses, whereas unguided media radiate wireless waves through space.

• **Topological Fault Tolerance:** Bus topologies suffer from single-cable backbone failure, whereas star topologies localize node failures to individual station drops connected to a central switch.

• **Connecting Device Intelligence:** Physical repeaters amplify raw signal voltages, data link switches forward frames by hardware MAC addresses, and network routers direct packets across subnets by IP address.

ESSENTIAL ARCHITECTURE & CORE SYNTAX CARDS

$$\text{Mesh Links} = \frac{n(n-1)}{2} \quad \text{(Full point-to-point redundancy)}$$

Fully Connected Mesh Topology Links Formula

$$\text{Bandwidth: Optical Fiber} \gg \text{Coaxial Cable} > \text{Twisted Pair (UTP)}$$

Transmission Media Bandwidth Hierarchy

$$\text{OSI 7 Layers} = \text{Phys} \rightarrow \text{DataLink} \rightarrow \text{Net} \rightarrow \text{Trans} \rightarrow \text{Sess} \rightarrow \text{Pres} \rightarrow \text{App}$$

ISO-OSI Reference Model Layer Hierarchy

$$\text{Switch (Layer 2 / MAC)} \quad \longleftrightarrow \quad \text{Router (Layer 3 / IP)}$$

Network Connecting Device Intelligence

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Distinguish Hub (broadcasts frames to all ports indiscriminately, causing collisions) from Switch (unicasts frames to destination port using an internal MAC table).

• **Alert:** In mesh topology calculations, a network of n nodes requires n(n - 1) divided by 2 duplex physical links and n - 1 ports per node.

• **Alert:** Optical fiber cables transmit signals via Total Internal Reflection (TIR) of light, rendering them 100 percent immune to electromagnetic interference (EMI).

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. What is a Computer Network? State two key advantages of networking computers. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Definition:** An interconnected collection of autonomous computers and devices linked through communication media to share hardware, software, and data resources. **[1 Mark]**

• **Advantages:** (1) Resource sharing (sharing expensive hardware like laser printers and storage); (2) Fast, reliable communication and centralized data management. **[1 Mark]**

2. Distinguish between a Hub and a Switch in computer networks. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Hub (Broadcasting):** A physical layer device that broadcasts incoming data packets to all connected ports indiscriminately, wasting bandwidth and causing traffic collisions. **[1 Mark]**

• **Switch (Intelligent Switching):** A data link layer device that inspects the destination hardware MAC address and forwards data frames strictly to the intended recipient port. **[1 Mark]**

3. Explain Star Topology and Bus Topology with diagrams, advantages, and disadvantages. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Star Topology:** All nodes connect individually to a central hub or switch. Advantage: single cable failure does not affect the rest of network. Disadvantage: failure of central switch disables entire network. **[1.5 Marks]**

• **Bus Topology:** All nodes share a single common linear backbone transmission cable with terminators at both ends. Advantage: minimal cabling and low cost. Disadvantage: fault in backbone cable breaks entire network. **[1.5 Marks]**

4. Compare guided media: Twisted Pair Cable, Coaxial Cable, and Optical Fiber Cable on transmission mechanism, bandwidth, and EMI susceptibility. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Twisted Pair Cable (UTP and STP):** Transmits electrical signals over twisted copper wires; moderate bandwidth; susceptible to electromagnetic interference (EMI); inexpensive. **[1 Mark]**

• **Coaxial Cable:** Solid central copper conductor surrounded by insulating layer and braided metal shield; higher bandwidth than twisted pair; good noise immunity. **[1 Mark]**

• **Optical Fiber Cable:** Transmits light pulses through ultra-pure glass or plastic cores via Total Internal Reflection; ultra-high bandwidth over long distances; completely immune to EMI. **[1 Mark]**

5. Classify computer networks into PAN, LAN, MAN, and WAN based on geographical span. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **PAN (Personal Area Network):** Spans a few meters around an individual user using Bluetooth or Wi-Fi (e.g. connecting phone to smartwatch). **[0.5 Mark]**

• **LAN (Local Area Network):** Confined to a single room, office floor, or school campus (up to 1 km), characterized by high data transfer speeds and private ownership. **[1 Mark]**

• **MAN (Metropolitan Area Network):** Covers an entire city or town (up to 50 km), such as municipal cable television networks. **[0.5 Mark]**

• **WAN (Wide Area Network):** Spans countries, continents, or the entire globe using satellites and undersea cables (e.g. the global Internet). **[1 Mark]**

6. A newly built school computer lab has 12 computers. Calculate how many physical cable links are required to connect them in (a) Star Topology, and (b) Fully Connected Mesh Topology. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **(a) Star Topology Calculation:** Each of the n computers requires one dedicated drop cable connecting to the central switch. Total links = n = 12 cable links. **[1.5 Marks]**

• **(b) Mesh Topology Calculation:** Formula for full mesh = n(n - 1) divided by 2. Substituting n = 12: Total links = (12 × 11) divided by 2 = $\frac{132}{2}$ = 66 physical links. **[1.5 Marks]**

7. Explain the role of Router and Gateway in connecting heterogeneous computer networks. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Router:** An intelligent network layer device that connects two or more different logical subnets, maintaining routing tables to forward IP data packets along the shortest, optimal path. **[1.5 Marks]**

• **Gateway:** A protocol converter that connects networks operating on fundamentally dissimilar communication architectures or protocol suites (e.g. connecting LAN to mainframe network). **[1.5 Marks]**